

CASE STUDY 2 - OFFER AMOUNT GENERATION FOR PERSONAL LOANS

Affordability, not just eligibility.

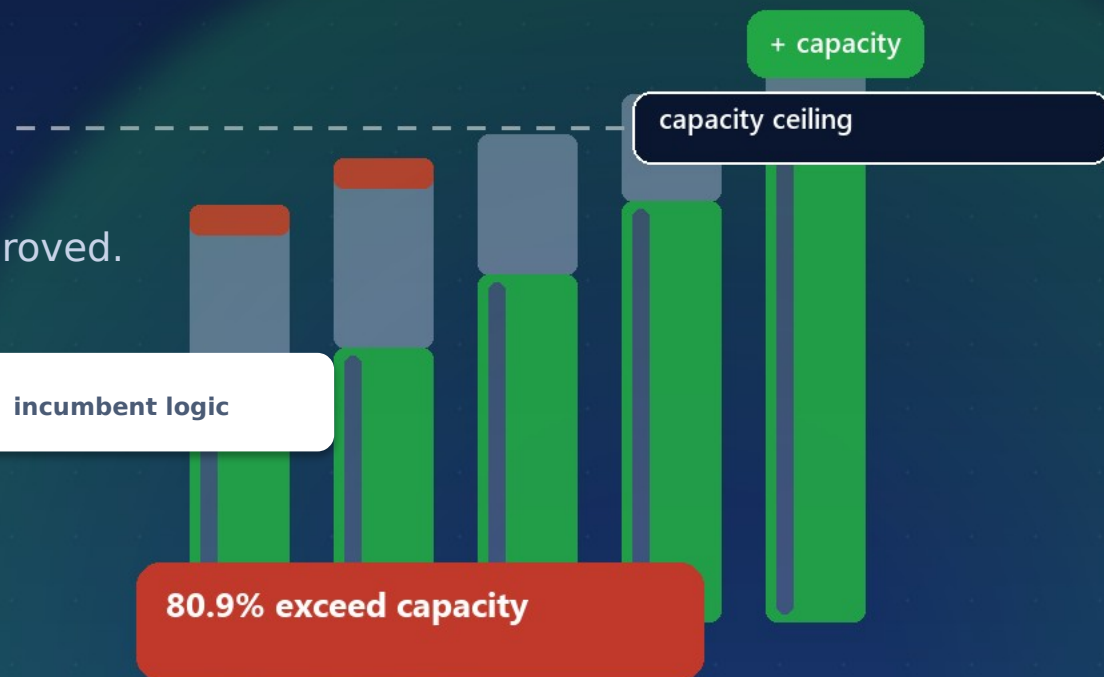
Re-sizing the personal-loan offer on what the customer has already proved.

**80.9% OF OFFERS EXCEED PROVEN
CAPACITY**

89.6 / 100 brief metric

60.0 incumbent logic

TVS Credit EPIC 8 · Analytics / Data Track



AFFORDABILITY, NOT JUST ELIGIBILITY.
Size the offer to what the customer has already proved.

1 DIAGNOSIS

2 COHORT SEGMENTATION

3 THE ENGINE

4 IMPACT & GUARDRAILS

The offer grid over-lends to customers who cannot carry it, and under-lends to customers who can.

TVS Credit sizes the personal-loan offer on eligibility bands. Measured against what each customer has demonstrably repaid, the grid is inverted, and both symptoms in the brief fall out of that one fact.

157,557

customer records

128

fields · 114 usable

10

fields barred by the data dictionary

16,621

build cohort (TEST · disbursed)

5,203

holdout cohort (VALIDATION · disbursed)

1.30%

base delinquency rate · 288 events

1 THE SYMPTOMS IN THE BRIEF

SYMPTOM 1 · “THE AMOUNT IS TOO LOW”

808 customers (627 in build) carry the remark “Require Higher Loan Amount”. Same demonstrated EMI capacity as the base: ₹31,652 vs ₹30,996. Offer 7% lower, 0% converted.

SYMPTOM 2 · THE LARGEST OFFERS LEAK

Offer Band	1	2	3	4	5	6
took TVS loan	1.1%	2.0%	4.6%	4.8%	5.6%	7.1%
borrowed elsewhere	18.4%	12.0%	12.3%	14.5%	14.5%	13.3%

Off-us take-up climbs 1.1% → 7.1% as the offer grows. At the top band, a third of everyone who borrowed, borrowed elsewhere.

2 THE DIAGNOSTIC · AFFORDABILITY COVERAGE

Offer ÷ (demonstrated EMI headroom × annuity factor 25.49), median by headroom quintile

Quintile	1	2	3	4	5	6
Coverage	2.57L	2.07L	1.73L	1.30L	1.20L	0.98L

80.9% of offers exceed proven capacity
99.9% in the lowest-capacity quintile, 25.5% in the highest. The grid is inverted.

3 WHAT THE OFFER RESPONDS TO

Average offer by bureau risk grade · a 2.6x spread

Risk Grade	1	2	3	4	5	6	7
Average Offer	2.57L	2.07L	1.73L	1.50L	1.20L	0.98L	1.09L

Largest prior personal loan

External bureau risk score

Total outstanding balance

Demonstrated EMI capacity

Total live EMI

Live EMI obligation

Bureau income proxy

0.328

0.314

0.282

0.166

0.151

0.139

0.039

It prices the past. It does not price the capacity.

4 COHORT SEGMENTATION · FOUR GROUPS THE CURRENT GRID CANNOT TELL APART

build cohort, n = 119,989

A

PROVEN, UNDER-OFFERED

14,904 · 12.4%

Avg offer ₹2,06,121 Off-us 6.5%

Headroom ₹17,524 30+ DPD 0.74%

Coverage 0.51x

RAISE Safest and most capable. Leaks hardest.

B

OVER-EXTENDED

34,721 · 28.9%

Avg offer ₹2,07,450 Off-us 4.7%

Headroom ₹2,000 30+ DPD 1.27%

Coverage 3.02x

CUT Same offer as A, 8.8x less capacity.

C

NO CAPACITY SIGNAL

42,105 · 35.1%

Avg offer ₹1,60,357 Off-us 3.1%

Headroom ₹0 30+ DPD 1.41%

Coverage -

HOLD, THEN INSTRUMENT No EMI history yet.

D

BROADLY ALIGNED

28,259 · 23.6%

Avg offer ₹1,53,931 Off-us 4.1%

Headroom ₹3,310 30+ DPD 1.44%

Coverage 1.53x

TRIM Mildly over-offered.

FACT (TVS DATA)	WHAT IT MEANS	WHAT TVS SHOULD DO
80.9% of offers exceed proven capacity; 3.89x median coverage in the lowest-capacity quintile.	The grid asks "is this customer eligible?" It never asks "what can this customer carry?"	Re-anchor on demonstrated EMI headroom, converted to principal at a stated annuity factor.
Cohorts A and B get the same offer on 8.8x different capacity. A leaks off-us at 6.5%; B runs 1.7x A's delinquency.	One number is doing two jobs badly. The safest customers are the ones walking.	Split the two with a risk score, then move rupees from B to A. Nothing new is spent.
808 customers asked for more, hold the same proven capacity as the base, and converted at 0.0%.	Serviceable demand is being turned away at the top of the capacity distribution.	Redistribute the existing book: raise where capacity is proven, cut where it is not.

SOURCE: epic_challenge_base_final_shared2.xlsx, 157,557 rows × 128 fields. Diagnostic on the build cohort BASE_TYPE=TEST, n = 119,989; coverage on the 77,884 customers with positive EMI headroom; segment delinquency on disbursed accounts only. CALCULATION (ours): coverage = offer ÷ (headroom × 25.49); cohorts defined as A coverage <1.0, B ≥2.0, D between, C no headroom. ASSUMPTION (ours): annuity factor at 24% p.a. over 36 months; every result holds across 20.28% p.a. and 24.48 months.

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1 DIAGNOSIS

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3 THE ENGINE

4 IMPACT & GUARDRAILS

Re-anchor on demonstrated capacity, modulate by risk, and move the rupees from B to A.

Same book in aggregate: ₹2,138 Cr to ₹2,169 Cr, +1.5%. Cohort A up 74%, cohort B down 21%. 89.6 out of 100 on the brief's metric against 60.0 for the incumbent logic.



2 ALGORITHM · SELECTED ON HOLDOUT, NOT ON CV

MODEL	CV AUC	HOLDOUT
LightGBM SELECTED	0.625 ±0.007	0.624
Logistic regression, L2	0.636 ±0.002	0.608
Random forest	0.630 ±0.006	0.587
Rank-average ensemble	0.648	0.617
Bureau score alone	0.563	0.564

TOP DRIVERS OF 30+ DPD IN MOB 1-6 (LightGBM split gain)

1 Median EMI

2 Age

3 Max credit amount

4 Total enquiry amount

5 External bureau score

6 Avg enquiry amount

7 Median enquiry amount

8 Credit amount

3 EVALUATION METRIC AND ITERATION LADDER

C1 Overall not lower	30%	1.004x	100 / 100
C2 Riskier offered less	40%	0.815x	74 / 100
C3 Converters not lower	30%	1.006x	100 / 100

C1 / C3: 100 if R ≥ 0.98, else 100 × max(0, 1 - (0.98 - R) ÷ 0.20) C2: 100 × min(1, (R - 0.98) ÷ 0.20)

89.6 WEIGHTED SCOREVS 60.0 Bootstrap 95% CI 78.1-99.3 (500 resamples)

4 HOLDOUT · RISK FALLS AS THE OFFER RISES

5,203 VALIDATION customers who took a PL. Policy fitted only on

Monotone, 2.9x gradient. Spearman ρ = -0.052, p = 0.00019. 72 events, so adjacent quintiles are NOT separable: the intervals say so.

5 REDISTRIBUTION · SAME BOOK, DIFFERENT SHAPE

Total book (Cr) ₹2,138→2,169
Net change +1.5%

Offered less 59.4%
Over capacity 52.5% → 41.8%

6 DECISION RULES · WHEN IT SAYS NOT YET

RAISE

TRIM

CUT

HOLD

FLOOR

Coverage < 1.0x and low risk to the capacity ceiling; the growth sits here
Coverage 1.0-2.0x toward capacity; tenure and pricing unchanged
Coverage > 2.0x, any risk band to the capacity ceiling; it was never serviceable
No demonstrated headroom at incumbent; says not yet, not no
Top risk decile floors at 0.30x whatever the capacity

7 EXPECTED BENEFIT

Convert the under-offered: 808 asked for more, same capacity as the base, 0% converted
Stop leaking large tickets: off-us take-up climbs 1.1% → 7.1% with offer size
Contain early delinquency: 18.5% smaller offers to 30+ DPD, at no cost to the average
Spend nothing extra: overall average 1.004x; the book is redistributed, not inflated

8 SENSITIVITY

multiplier grid: score 49.6-89.6, median 77.9
pricing: 86.7-93.4 across every scenario

ROLLOUT

M1-2

M3-4

M5

M6

Freeze barred-field list; ship the capacity feature
Instrument coverage and off-us leakage as standing KPIs
Champion/challenger on 20% of the pre-approved base
Extend to cohort C via consented cash-flow data

WHAT WE WILL NOT CLAIM

- 216 training events. Holdout AUC 0.624 is honest, not impressive: every claim is sized to it.
- Adjacent quintiles are not separable on 72 events. We show the intervals rather than hide them.
- The risk model is fitted on the 13.9% who took a loan. Generalising to non-takers is untested.
- 40.6% of customers show no demonstrated EMI headroom and fall back to the incumbent offer, unchanged.